

Product Requirements Document (PRD)

CliFactory: Terminal-Based Industrial Tycoon Game

1. Problem Statement & Description

CliFactory is a terminal-based industrial tycoon game written in Python, engineered using a clean Functional Core / Imperative Shell architecture. The application manages a stateful industrial production and supply chain system over a monthly turn-based cycle. It tracks capital reserves, raw material inventories, finished goods, dynamic maintenance overhead, progressive factory scale tiers, multi-tier R&D tech trees, risky exploration sectors, corporate futures contracts, tactical minor actions, and a dynamic 40-event pool. Strict adherence to clean data separation, risk modulation, and robust input validation is maintained.

2. Interface & Exact Formatting Rules

- **Run Command:** `python main.py`
- **Divider Line:** Section breaks strictly use 75 characters of equals signs (=) or hyphens (-).
- **Dashboard Display:** Renders the active difficulty tier, current month, capital, raw materials, finished goods, monthly upkeep, max sell limits, active R&D tiers, and remaining major/minor action counters.
- **Risk Transparency:** Minor action menus dynamically display the current calculated risk percentage (e.g., `Risk: 15%`) for each category rather than labeling individual choices as "Positive" or "Risky" to encourage player deduction.

3. Functional Requirements

FR 1: Turn-Based Action System & UI

- **Major Actions (Max 1/mo):** Prospect Unknown Sector, Construct Factory Expansion, Invest in R&D, Corporate Futures Contract, Bulk Material Import.
- **Minor Actions (Max 2/mo):** 25 specific Tactical Minor Actions across 5 categories (Operations, Maintenance, Market, Logistics, Human Capital) or Secure Sales Contracts.
 - *Note:* Actions like Route Optimization apply permanent stacking buffs (+\$2 market price).
- **Action Exhaustion & Progression:** The menu dynamically hides unavailable options and replaces text with an explicit `[UNAVAILABLE]` tag immediately once quotas are reached.

When all actions are spent, option 8 dynamically updates to **ALL ACTIONS EXHAUSTED - Proceed to Next Month.**

FR 2: Economic & Risk Mechanics

- **Passive Loop & Itemized Logs:** End-of-month conversion of raw materials into finished goods. The rollover MUST print an explicit, itemized summary log detailing all maintenance deductions, extracted materials, refined goods, active event triggers, and quota penalties/bonuses.
- **Minor Action Probabilities:** "Positive" tactical minor actions guarantee 100% success. "Risky" minor actions roll against the player's active risk percentage.
- **Centralized Risk Bounds:** All probabilistic risks are managed by a centralized helper function (`calculate_effective_risk`) that enforces a strict 5% floor and a 100% ceiling. Risk is dynamically reduced by Safety R&D and increased by ruthless expansions.
- **Corporate Futures Contracts:** Sets a monthly quota (40 units) in exchange for a 2.5x price multiplier. Failing the quota triggers a 10% penalty on total capital (calculated dynamically after the month's sales revenue is added). Enforces a strict 3-month cooldown.

FR 3: Monthly Events System

- **Trigger Mechanism:** At the start of a new month, there is a 50% base chance for a random event to occur.
- **Event Pool:** 40 distinct events (20 Positive, 20 Negative) spanning government subsidies, strikes, lawsuits, and innovations.
- **Dynamic Mitigation & Choices:** The likelihood of rolling a negative event versus a positive one is mitigated by the player's Safety level. Negative events feature interactive binary choices (Strictly 1 or 2) to mitigate damage or negotiate penalties. No instant-loss events exist.

FR 4: Research & Development (R&D)

- **Tech Fields:** Logistics (upkeep/sales), Efficiency (passive mining/refining), Safety (risk reduction), Marketing (bonus profit quotas).
- **Tier Limits:** Each field has a maximum capacity of Tier 4. Investing past Tier 4 is rejected and refunds the action slot.

FR 5: Progression & Victory Conditions

- **Factory Tiers:** Factory → Large Factory → Large Factory Complex → Mega Factory → Mega Factory Supercomplex → Ultra-Industrial Zenith Complex.
- **Win Conditions by Tier:**
 - **Subsidized Startup (Easy):** Reach \$5,000 Capital and 50 goods/mo output rate.

- **Mid-Market (Normal):** Reach \$15,000 Capital, 120 goods/mo output rate, and max out 2 R&D fields to Tier 4.
- **Hostile Takeover (Hard):** Reach \$35,000 Capital, 250 goods/mo output rate, max out all 4 R&D fields to Tier 4, and fulfill 3 futures contracts.

4. Edge Cases & Verbatim Handling

#	Situation	Expected Handling / Output
EC1	Out-of-bounds menu selection	Print <code>[!] Invalid choice. Press Enter to try again.</code>
EC2	Capital falls to or below \$0 mid-action	Immediate dashboard evaluation triggers <code>GAME OVER: BANKRUPTCY</code> without waiting for the end-of-month rollover.
EC3	Actions selected when quota exhausted	Print explicit lock notice and return to menu.
EC4	Futures selected during cooldown	Display remaining months and refund the major action slot.
EC5	Invalid non-numeric input on binary event choice	Strict input loop rejects letters/symbols; re-prompts with <code>[!] Invalid choice. Please enter 1 or 2.</code>
EC6	Risk percentage stacks via expansions	Centralized calculator forces a ceiling of 1.0 (100%) and a floor of 0.05 (5%).
EC7	R&D investment past Tier 4	Prints warning that technologies are fully researched and refunds the major action slot.

5. Test Cases (EP & BVA)

Functional Requirements (FR) & Logic Testing

ID	Requirement	Scenario (EP/BVA)	Steps & Input	Expected Result
TC_FR1_01	FR 1 ▾	BVA (Minor Action Limits) ▾	Execute exactly 2 Minor Actions via Option 6 or 7.	Dashboard renders Minor actions: 0/2. Options 6 and 7 text changes to [UNAVAILABLE] . Selecting them prints rejection.
TC_FR1_02	FR 1 ▾	BVA (Major Action Limit) ▾	Execute exactly 1 Major Action via Option 1.	Options 1-5 marked [UNAVAILABLE] . Trying to trigger Option 2 explicitly rejects the action.
TC_FR1_03	FR 1 ▾	EP (Action Exhaustion Chec... ▾	Complete 1 Major and 2 Minor Actions.	Option 8 text dynamically updates to "ALL ACTIONS EXHAUSTED - Proceed to Next Month".
TC_FR2_01	FR 2 ▾	BVA (Risk Floor Clamp) ▾	Stack Safety-First expansions and Safety R&D until risk reduction mathematically passes below 0.0.	calculate_effective_risk clamps value. UI displays Risk: 5%. Internal negative event logic uses 0.05 baseline.
TC_FR2_02	FR 2 ▾	BVA (Risk Ceiling Clamp) ▾	Stack Ruthless and Cost-Cutting expansions until risk modifier mathematically exceeds 1.0.	calculate_effective_risk clamps value. UI displays Risk: 100%. Minor risky actions trigger negative outcome 100% of the time.
TC_FR2_03	FR 2 ▾	BVA (Futures Quota Miss) ▾	Activate Futures Contract. End month with exactly 39 total goods sold.	End of month itemized log prints: "Futures Contract: Sold only 39/40 goods." Deducts exactly 10% of total capital balance.
TC_FR2_04	FR 2 ▾	BVA (Futures Quota Pass) ▾	Activate Futures Contract. End month with exactly 40 total goods sold.	End of month itemized log prints: "Futures Contract: Quota fulfilled successfully!" No penalty applied.

ID	Requirement	Scenario (EP/BVA)	Steps & Input	Expected Result
TC_FR4_01	FR 4 ▾	BVA (R&D Tier Max Limit) ▾	Select R&D (Option 3), invest in Logistics 5 times.	5th attempt explicitly rejects the upgrade, states "technologies fully researched," and refunds the Major Action slot and the \$100.
TC_FR5_01	FR 5 ▾	BVA (Victory Threshold) ▾	On Easy, reach exactly \$4999 Capital and 50 output.	Game continues. Dashboard refreshes for next month.
TC_FR5_02	FR 5 ▾	BVA (Victory Threshold) ▾	On Easy, reach exactly \$5000 Capital and 50 output.	Next dashboard loop evaluates True; instantly clears screen and displays VICTORY: CORPORATE HEGEMONY .

Edge Cases (EC) & Input Validation Testing

ID	Edge Case	Scenario (EP/BVA)	Steps & Input	Expected Verbatim Output / Result
TC_EC1_01	EC1 ▾	BVA (Menu Number Bounds)	At main menu, enter 0, then 10.	[!] Invalid choice. Press Enter to try again. Menu redraws safely without crashing.
TC_EC1_02	EC1 ▾	EP (Menu Invalid Type)	At main menu, enter A, !, or leave blank.	[!] Invalid choice. Press Enter to try again.
TC_EC2_01	EC2 ▾	BVA (Exact \$0 Bankruptcy)	Capital is \$200. Trigger Bulk Import (-\$200). Capital reaches exactly \$0.	Loop immediately catches state; triggers GAME OVER: BANKRUPTCY without waiting for the month to end.
TC_EC2_02	EC2 ▾	BVA (Negative Bankruptcy)	Capital is \$30. Monthly event triggers -\$80 fine. Capital reaches -\$50.	Loop immediately catches state; triggers GAME OVER: BANKRUPTCY without waiting for month to end.
TC_EC4_01	EC4 ▾	EP (Cooldown Enforcement)	Trigger Futures Contract. Next month, try to trigger again.	Rejects action. Prints [!] Futures contract is on

ID	Edge Case	Scenario (EP/BVA)	Steps & Input	Expected Verbatim Output / Result
				cooldown! (2 months remaining). Action slot is refunded.
TC_EC5_01	EC5 ▾	EP (Binary Event Type Loop)	Event prompts for 1/2 choice. Input 3, A, or empty string.	Prompts: [!] Invalid choice. Please enter 1 or 2. Forces user to stay in strict loop until valid 1 or 2 is submitted.